

1.0 Design Parameters Exploration

In the report, we explored the effects of wire length, RC values, and switching block patterns on the delay and area cost of routing.

1.1 Wire Length Effect on Routing parameters

The geometric averages across 8 benchmarks are recorded in the table by varying the wire lengths in the XML file. In addition, each benchmark's results are processed and averaged across 5 different seed values to reduce CAD noise.

length	Minimum channel width	Low-stress routing area per tile	low-stress critical path vs wire-length
1	52	7847	8.442671
2	53	5845	7.017566
4	62	5271	6.880281
8	94	5980	8.095856
16	172	8064	9.270965

Table 1: Effect of Wire length on Geometric Averages of benchmarks

The plots of critical path delay vs. length and average area per tile vs. length are shown below. It is found that the critical path delay and low-stress routing area per tile are lowest when the wire length is 2 or 4. For a wire length of 1, although it provides finer control over routing paths due to the increased flexibility in connecting adjacent blocks, it requires more intermediate switches between wire segments. This is because the limited switch box size necessitates frequent transitions between short wire segments. As a result, the number of routing switches increases, leading to higher delay and area overhead. Longer wires reduce the number of switch connections, which lowers the overhead from switches. However, they also introduce greater delay due to the increased resistance and capacitance (RC delay) associated with longer wires.

Wire lengths of 4 and 8 provide the best balance between these trade-offs. They reduce the number of switches required for routing while keeping the RC delay manageable. As a result, they offer an optimal combination of area efficiency and delay performance.

As the wire length gets longer, the minimum channel width increases. This suggests that When wire length increases, fewer wires are available to connect all possible source-destination pairs. This creates more demand for the available wires, requiring additional wires to handle the increased routing load.

In conclusion, the best routing wire length would ideally balance both the low-stress routing area per tile and critical path delay. Based on the data:

Wire Length 4 shows the lowest area and critical path delay.

Wire Length 2 also performs well in terms of area and delay, with slight overhead compared

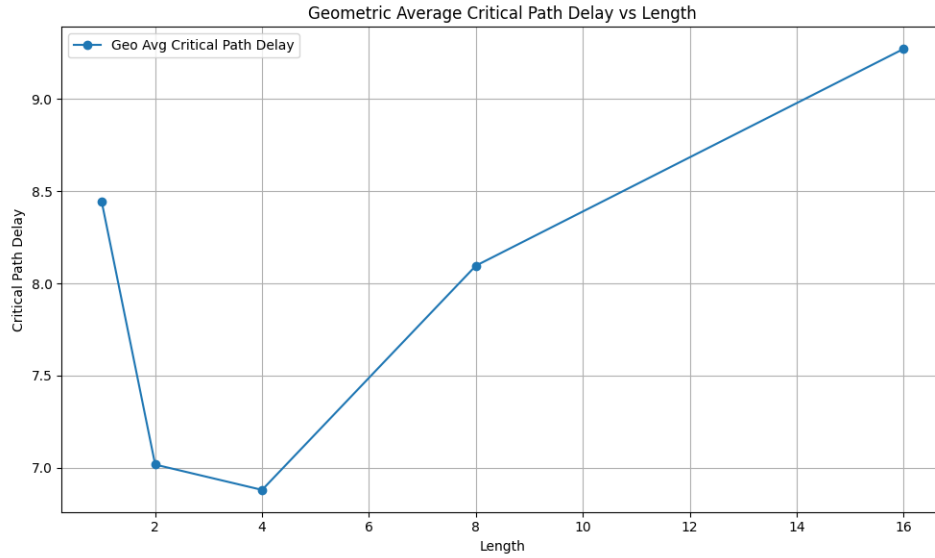


Figure 1: Critical Path Delay Vs. Wire length

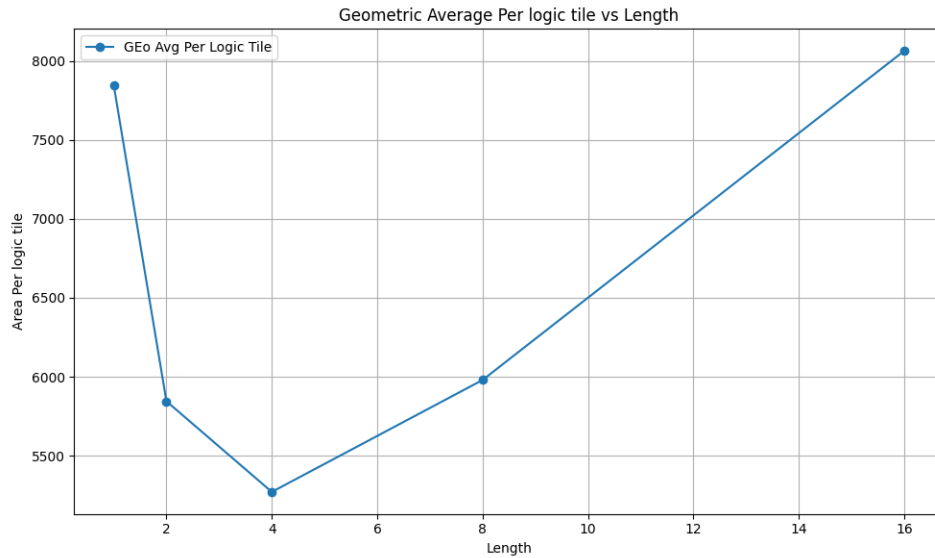


Figure 2: low-stress routing area per tile Vs. Wire length

to wire Length 4.

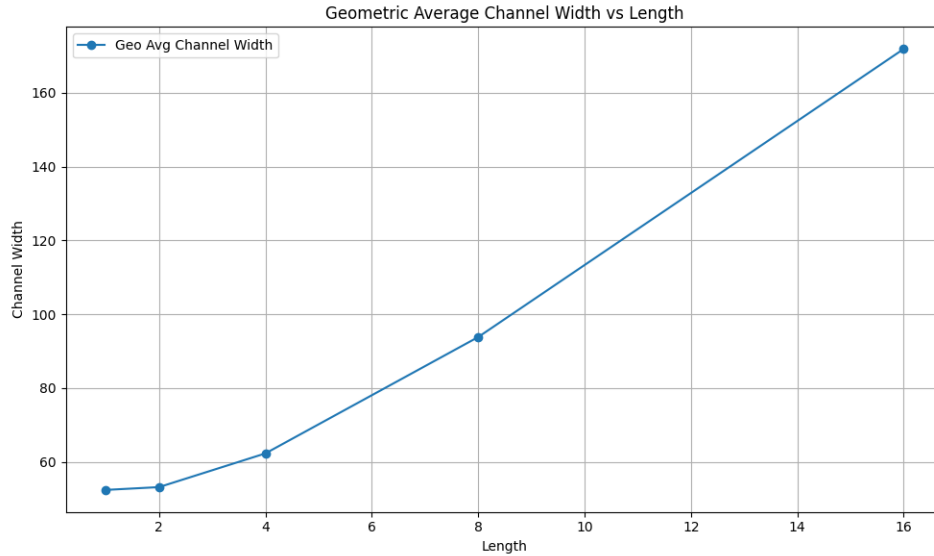


Figure 3: Minimum Channel Width Vs. Wire length

1.2 Block to routing connectivity

In k6_N10_40nm.xml, we explore the ratio of input and output pins having access to a limited portion of available tracks at wire length 4 and explore its effect on critical path delay and area cost.

It is found that low F_{in} and F_{out} ratios lead to higher minimum channel widths because only a small portion of the routing tracks is accessible for connections. This limited flexibility requires the existing routing tracks to be utilized more efficiently, thus leading to a higher channel width. However, this configuration also results in a lower low-stress area per tile, as fewer connections reduce the demand for switch box usage and interconnect fabric. As the ratios get higher, channel width decreases correspondingly, reflecting improved flexibility on routing. Minimum channel width saturates at 57 when F_{in} and F_{out} reach 0.5 and higher. In the compensation, the routing area grows with a higher ratio, as higher connectivity requires more switches and interconnect fabric to support the increased pin-to-track options. The delay also increases slightly (7.88s) as higher pin connectivity adds parasitics delays to the routing switches, which slows down signal propagation.

Similar test cases were conducted using the sparse crossbar architecture, and the results reveal a similar trend to the full crossbar. However, the minimum channel width is slightly smaller in the sparse crossbar architecture. This reduction is because the sparse crossbar shifts some of the routing burden from global routing to local routing, leveraging logical equivalence to reduce pressure on global routing resources.

On the other hand, the routing area is generally higher for the sparse crossbar compared to the full crossbar. Although the sparse crossbar uses fewer switches internally, it compensates for the reduced crossbar connectivity by requiring additional routing tracks and interconnect

F _{in} =F _{out}	Minimum channel width	Low-stress routing area per tile	low-stress critical path vs wire-length
0.15	60	6165	7.13
0.5	57	8160	7.80
1.0	57	8569	7.88

Table 2: Output Results by varying Fin, Fout in k6_N10_40nm.xml

resources to maintain routing flexibility, which increases the overall area usage.

The critical path delay is higher in the sparse crossbar architecture. This is because the reduced local routing flexibility forces the router to create more complex paths, and thus increases delays. The limited number of local routing options in the sparse crossbar introduces challenges in achieving optimal routing, ultimately leading to higher delays.

F _{in} =F _{out}	Minimum channel width	Low-stress routing area	low-stress critical path vs wire-length
0.15	61	8293	7.17
0.5	54	8859	7.81
1.0	54	9454	8.09

Table 3: Output Results by varying Fin, Fout in k6_N10_sparse_crossbar_40nm.xml

Overall, both options suggest $F_{in}=F_{out}=0.5$ and above appear to yield the smallest minimum channel width. As F_c reach higher, the effect of minimum channel width shows diminishing returns with minimum improvement. In terms of delay and average routing area per tile, $F_c = 0.15$ seems to be the best overall choice for achieving lower routing area and critical path delay.

2.0 Optimization Study

Through previous research, it is found that wire lengths 2 and 4, and $F_c=0.15$ have the best area-delay product. Therefore, the following optimization study would primarily focus on implementing lengths of 2 and 4, while exploring other F_c values around 0.15. First, we explore the effect of mixing different wire lengths, it is found that mixing wire lengths of 2 and 4 yields although they yield a smaller minimum channel width, they still have a worse area-delay product than the results found in parts 1 and 2. The main reason the area-delay product increases when using both length 4 and length 2 wires is likely due to increased routing complexity and inefficiencies in wire utilization. Mixing wire lengths complicates the routing process, leading to longer and more convoluted signal paths, more switches, and

potentially more congestion. This results in higher area usage and increased signal delay. In contrast, using only one wire length simplifies the routing task, allowing the router to optimize the layout more effectively, which can lead to a lower area-delay product.

Freq_len4: Freq_len2	Minimum channel width	Low-stress routing area	low-stress critical path
0.9:0.1	43	7078	7.05
0.65:0.35	42	7176	6.92
0.35:0.65	40	7577	6.93

Table 4: Output Results by varying wire length freq, $F_{out}=F_{in}=0.5$, $sw=Wilton$ in k6_N10_40nm.xml

Also, by varying R_{metal} and C_{metal} in wire-length4, while keeping other parameters the same, it is found that doubling R_{metal} and 40% less C_{metal} yield the best area-delay product.

RC ratio	Minimum channel width	Low-stress routing area	low-stress critical path
1:1	51	5362	6.88
2:0.6	51	5348	6.67
0.4:1.2	52	5414	6.81

Table 5: Output Results by RC metal Ratio freq, $F_{out}=F_{in}=0.5$, $sw=Wilton$, wire.length=4 in k6_N10_40nm.xml

In addition, we can place 15% of the length of 4 metal wires on a higher level to reduce the R_{metal} values, providing a slightly better result. The best area-product Is found to be:

85% Length 4 wire with $R_{metal}=202$, $C_{metal}=13.5e-15$, $sw\ pattern = 1\ 0\ 1\ 0\ 1$

15% Length 4 wire with $R_{metal}=25.25$, $C_{metal}=22.5e-15$, $sw\ pattern = 1\ 0\ 1\ 0\ 1$

$F_c=0.15$

Low-stress channel width	low-stress routing area	low-stress critical path
57	5041	6.79

Table 6: Optimized area-delay product in k6_N10_40nm.xml

This architecture works well because it strategically balances routing complexity and maximizes resource utilization. By focusing primarily on length 4 wires and adjusting the parameters accordingly, the design minimizes the inefficiencies seen with mixed wire lengths while

achieving excellent area-delay trade-offs. In addition, adjusting resistance and capacitance on metal wires ensures faster signal propagation without compromising area efficiency. The resulting design not only achieves a lower area-delay product but also balances channel width, critical path delay, and area, making it highly effective for the given routing architecture.

3.0 Appendix

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<architecture>

<models>
</models>

<tiles>
  <tile name="io" area="0">
    <sub_tile name="io" capacity="8">
      <equivalent_sites>
        <site pb_type="io" pin_mapping="direct" />
      </equivalent_sites>
      <input name="outpad" num_pins="1" />
      <output name="inpad" num_pins="1" />
      <clock name="clock" num_pins="1" />
      <fc in_type="frac" in_val="0.15" out_type="frac" out_val="0.15" />

      <pinlocations pattern="custom">
        <loc side="left">io.outpad io.inpad io.clock</loc>
        <loc side="top">io.outpad io.inpad io.clock</loc>
        <loc side="right">io.outpad io.inpad io.clock</loc>
        <loc side="bottom">io.outpad io.inpad io.clock</loc>
      </pinlocations>
    </sub_tile>
  </tile>

  <tile name="clb" area="18000">

    <sub_tile name="clb">
      <equivalent_sites>
        <site pb_type="clb" pin_mapping="direct" />
      </equivalent_sites>

      <input name="I" num_pins="40" equivalent="full" />
      <output name="O" num_pins="10" equivalent="instance" />
      <clock name="clk" num_pins="1" />
    </sub_tile>
  </tile>
</tiles>
```

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        <fc in_type="frac" in_val="0.15" out_type="frac" out_val="0.15" />
        <pinlocations pattern="spread" />
    </sub_tile>
</tile>
</tiles>

<layout>
    <auto_layout aspect_ratio="1.0">

        <perimeter type="io" priority="100" />
        <corners type="EMPTY" priority="101" />

        <fill type="clb" priority="10" />
    </auto_layout>
</layout>

<device>

    <sizing R_minW_nmos="8926" R_minW_pmos="16067" />

    <area grid_logic_tile_area="0" />
    <chan_width_distr>
        <x distr="uniform" peak="1.2" />
        <y distr="uniform" peak="1.2" />
    </chan_width_distr>

    <switch_block type="wilton" fs="3" />

    <connection_block input_switch_name="ipin_cblock" />
</device>
<switchlist>

    <switch type="mux" name="0" R="551" Cin=".77e-15" Cout="4e-15" Tdel="58e-12"
    mux_trans_size="2.630740" buf_size="27.645901" />

    <switch type="mux" name="ipin_cblock" R="2231.5" Cout="0." Cin="1.47e-15"
    Tdel="7.247000e-11" mux_trans_size="1.222260" buf_size="auto" />
</switchlist>

<segmentlist>

    <segment name="L4_low" freq="0.75" length="4" type="unidir" Rmetal="202"
    Cmetal="13.5e-15">
        <mux name="0" />

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    <sb type="pattern">1 0 1 0 1</sb>
    <cb type="pattern">1 1 1 1</cb>
</segment>

<segment name="L4_high" freq = "0.15" length="4" type="unidir" Rmetal="25.25"
Cmetal="22.5e-15">
    <mux name="0" />
    <sb type="pattern">1 0 1 0 1</sb>
    <cb type="pattern">1 1 1 1</cb>
</segment>

<!-- <segment freq="1" freq="0.1" length="2" type="unidir" Rmetal="202" Cmetal="13.5e
<mux name="0" />
<sb type="pattern">1 0 1</sb>
<cb type="pattern">1 1</cb>
</segment>
-->
</segmentlist>

<complexblocklist>

<pb_type name="io">
    <input name="outpad" num_pins="1" />
    <output name="inpad" num_pins="1" />
    <clock name="clock" num_pins="1" />

    <mode name="inpad">
        <pb_type name="inpad" blif_model=".input" num_pb="1">
            <output name="inpad" num_pins="1" />
        </pb_type>
        <interconnect>
            <direct name="inpad" input="inpad.inpad" output="io.inpad">
                <delay_constant max="4.243e-11" in_port="inpad.inpad" out_port="io.inpad" />
            </direct>
        </interconnect>
    </mode>
    <mode name="outpad">
        <pb_type name="outpad" blif_model=".output" num_pb="1">
            <input name="outpad" num_pins="1" />
        </pb_type>
        <interconnect>
            <direct name="outpad" input="io.outpad" output="outpad.outpad">
                <delay_constant max="1.394e-11" in_port="io.outpad" out_port="outpad.outpad" />
            </direct>
        </interconnect>
    </mode>

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    <power method="ignore" />
</pb_type>

<pb_type name="clb">
  <input name="I" num_pins="40" equivalent="full" />
  <output name="O" num_pins="10" equivalent="instance" />
  <clock name="clk" num_pins="1" />

  <pb_type name="fle" num_pb="10">
    <input name="in" num_pins="6" />
    <output name="out" num_pins="1" />
    <clock name="clk" num_pins="1" />

    <mode name="n1_lut6">

      <pb_type name="ble6" num_pb="1">
        <input name="in" num_pins="6" />
        <output name="out" num_pins="1" />
        <clock name="clk" num_pins="1" />

        <pb_type name="lut6" blif_model=".names" num_pb="1" class="lut">
          <input name="in" num_pins="6" port_class="lut_in" />
          <output name="out" num_pins="1" port_class="lut_out" />

          <delay_matrix type="max" in_port="lut6.in" out_port="lut6.out">
            82e-12
            173e-12
            261e-12
            263e-12
            398e-12
            397e-12
          </delay_matrix>
        </pb_type>

        <pb_type name="ff" blif_model=".latch" num_pb="1" class="flipflop">
          <input name="D" num_pins="1" port_class="D" />
          <output name="Q" num_pins="1" port_class="Q" />
          <clock name="clk" num_pins="1" port_class="clock" />
          <T_setup value="66e-12" port="ff.D" clock="clk" />
          <T_clock_to_Q max="124e-12" port="ff.Q" clock="clk" />
        </pb_type>

        <interconnect>
          <direct name="direct1" input="ble6.in" output="lut6[0:0].in" />

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        <direct name="direct2" input="lut6.out" output="ff.D">
            <pack_pattern name="ble6" in_port="lut6.out" out_port="ff.D" />
        </direct>
        <direct name="direct3" input="ble6.clk" output="ff.clk" />
        <mux name="mux1" input="ff.Q-lut6.out" output="ble6.out">
            <delay_constant max="25e-12" in_port="lut6.out" out_port="ble6.out" />
            <delay_constant max="45e-12" in_port="ff.Q" out_port="ble6.out" />
        </mux>
    </interconnect>
</pb_type>
<interconnect>
    <direct name="direct1" input="fle.in" output="ble6.in" />
    <direct name="direct2" input="ble6.out" output="fle.out[0:0]" />
    <direct name="direct3" input="fle.clk" output="ble6.clk" />
</interconnect>
</mode>

</pb_type>
<interconnect>

    <complete name="crossbar" input="clb.I-fle[9:0].out" output="fle[9:0].in">
        <delay_constant max="95e-12" in_port="clb.I" out_port="fle[9:0].in" />
        <delay_constant max="75e-12" in_port="fle[9:0].out" out_port="fle[9:0].in" />
    </complete>
    <complete name="clks" input="clb.clk" output="fle[9:0].clk">
    </complete>

    <direct name="clbouts1" input="fle[9:0].out" output="clb.O" />
</interconnect>
</pb_type>

</complexblocklist>
<power>
    <local_interconnect C_wire="2.5e-10" />
    <mux_transistor_size mux_transistor_size="3" />
    <FF_size FF_size="4" />
    <LUT_transistor_size LUT_transistor_size="4" />
</power>
<clocks>
    <clock buffer_size="auto" C_wire="2.5e-10" />
</clocks>
</architecture>

```